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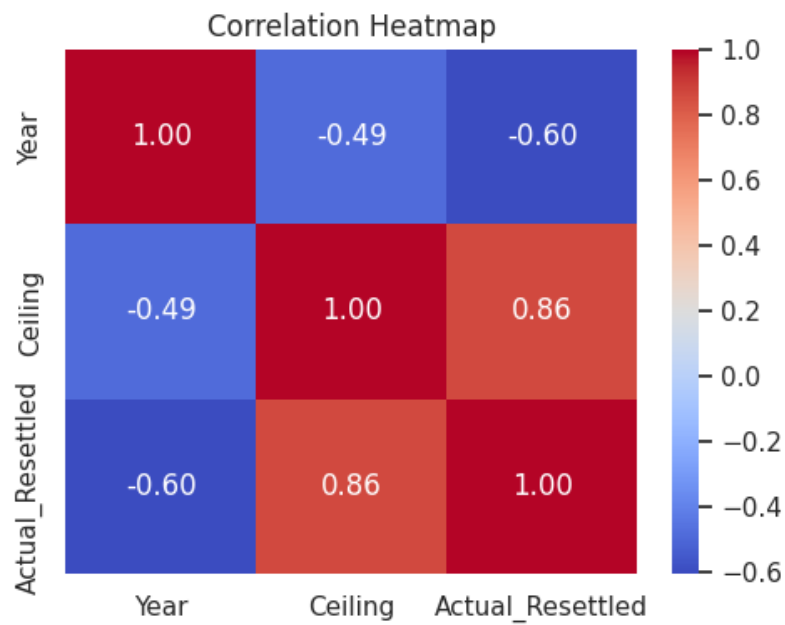
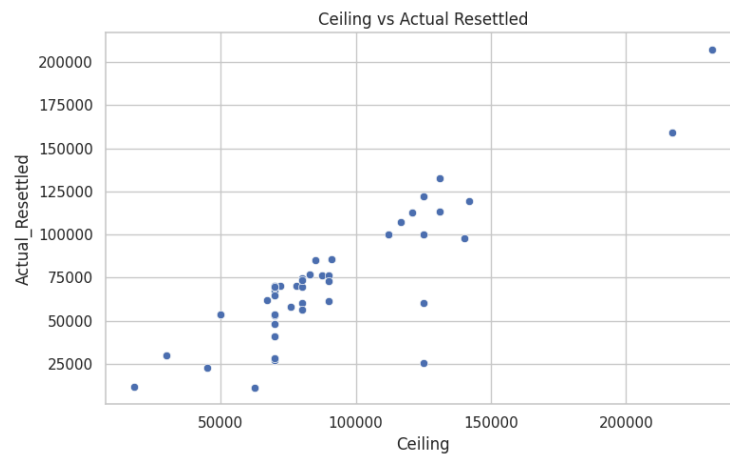
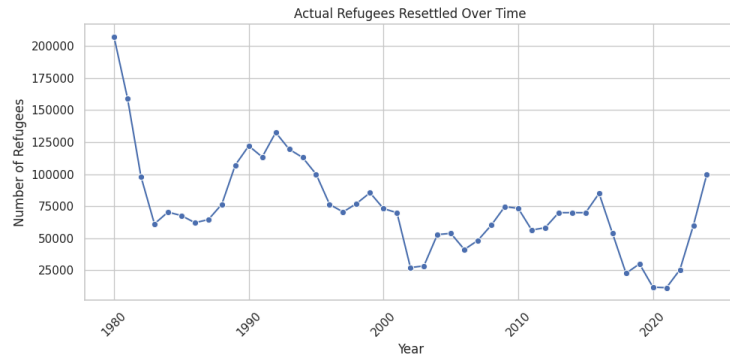
Final

Due: May 2, 2025

Refugee Resettlement

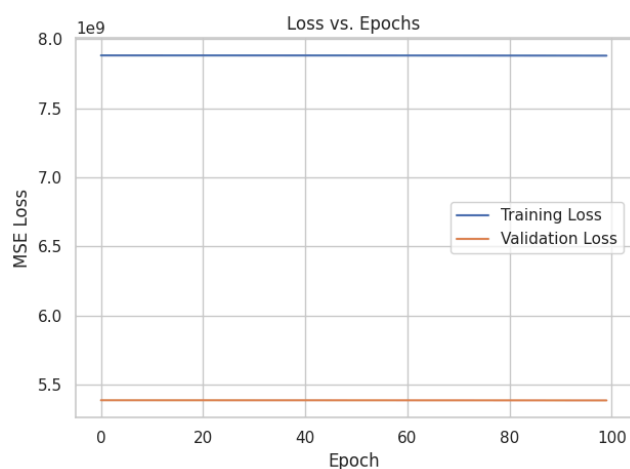
In this final assignment, I chose a dataset of refugee resettlement because I wanted to analyze the trends of resettlement in the United States. I used a real-world dataset from the U.S. Department, and in this assignment, the aim was to model and predict the number of actual refugee resettlements based on the ceiling and year. We were tasked with creating three models in the course, including supervised models, cross-validation, and regression analysis. With these models, the aim was to evaluate and compare how well the machine learning models did in forecasting refugee resettlement trends.

In this assignment, I had to clean the data in this Excel file because null values and non-numeric entries existed. The columns were also converted so that we could see the proper data type. In our visualizations, we mainly look at a line plot when it comes to looking at the resettlements over time, and a scatterplot when comparing the amount resettled vs the actual amount. The heatmap was included to show the relations among the year, the ceiling, and those resettled. For our model selections, a basic model was a linear regression model because it is simple and easy to read when it comes to data like this. Next was a decision tree where we tried to capture any non-linear relationships. The third model was a neural network (MLP) that was trained with 2 hidden layers and a multi-layer perceptron. We trained the model for 100 epochs, and each epoch was different.



To evaluate model performance, we applied both standard regression metrics and cross-validation techniques. For each of the three models — Linear Regression, Decision Tree, and Neural Network (MLP) — we calculated the Mean Squared Error (MSE) and R-squared (R^2) score on the test set. The Linear Regression model achieved an MSE of approximately **341,130,611.90** and an R^2 score of **0.6729**, suggesting moderate predictive performance. The Decision Tree Regressor, with a depth of 5, performed best overall on the test set with an MSE of **232,564,734.61** and an R^2 score of **0.7770**, indicating it captured non-linear patterns more effectively. The Neural Network, despite being trained over 100 epochs with two hidden layers, yielded a high test MSE of **4.7 billion** and a negative R^2 of **-3.5248**, which points to big overfitting or poor generalization. We also conducted 5-fold cross-validation to estimate how each model would perform on unseen data. Linear Regression had an average R^2 of **-0.3041** and average MSE of **477,829,655.65**, while the Decision Tree had an even lower average R^2 of **-1.2801** and MSE of **1.2 billion**, indicating instability across folds and possible sensitivity to small training subsets. These metrics, alongside plotted learning curves and loss over epochs, were used to compare the models' bias and variance trade-offs. The Decision Tree offered the best balance for this dataset, while the Neural Network failed to generalize due to the limited dataset size and lack of feature richness.

To improve model performance and address issues such as overfitting and poor generalization, we explored model optimization techniques including regularization



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hyperparameter tuning. Although Ridge Regression (a regularized form of linear regression) was not covered deeply in class, it was briefly explored for comparison. Using 5-fold cross-validation across various regularization strengths (alpha values from 0.01 to 100), we observed that higher alpha values marginally improved the R^2 score. The best performance was achieved with **alpha = 100**, yielding an average MSE of approximately **473,130,300** and an R^2 score of **-0.2929**, slightly better than the baseline linear model. However, the improvement was not substantial, suggesting that regularization alone does not significantly enhance performance for this dataset. In contrast, hyperparameter tuning was more impactful for the Decision Tree model. Using a grid search over parameters such as max_depth and min_samples, we identified max_depth of 8 and min_samples of 2 as the best configuration, reducing the cross-validated MSE to **966,330,812.53**. This tuning helped stabilize model performance across different data splits. Overall, the optimization step highlighted the importance of tailoring model complexity to the size and structure of the dataset, with simpler, well-tuned models often outperforming more complex ones like neural networks in small, tabular data settings.

This project aimed to predict actual refugee resettlement numbers in the U.S. based on historical ceilings and years, using machine learning techniques. After cleaning and preprocessing a real-world dataset from the U.S. Department of State, we visualized trends over time and investigated correlations among features. We implemented three models — Linear Regression, Decision Tree, and a Neural Network (MLP) — to evaluate their effectiveness on this regression task. The Decision Tree model, particularly after tuning, consistently outperformed others with an R^2 score of **0.7770**, suggesting it effectively captured the non-linear patterns in the data. Linear Regression, though simple, provided interpretable results with moderate performance, while the Neural Network struggled due to limited data and overfitting,

as seen in its high MSE and negative R^2 . The use of cross-validation revealed model instability and reinforced the importance of tuning and model selection. Visual tools like learning curves, scatter plots, and epoch loss graphs helped illustrate these results clearly. While the project demonstrated the potential of machine learning in understanding policy-driven trends like refugee resettlement, it also revealed the limitations of using small datasets for complex models. In future work, incorporating more features, such as country of origin, political factors, or economic indicators, could improve model robustness and generalizability.

